

Health, Risk, And Labor Markets

Core Frameworks And Measurement

Special Topic 6: Labor Market, Health, and Population – Week 1

Central question

- How should labor economists model and measure health when health affects both productivity and the feasible set for work?
- Health is not only a control variable. It changes work capacity, job choice, productivity, mobility, wages, amenities, and welfare.
- Week 1 discipline: name the health object before interpreting a health coefficient.

Why this is labor economics

- Health changes whether work is feasible and how much work is sustainable.
- Health changes which jobs, schedules, commutes, risks, and amenities are tolerable.
- Health changes productivity inside jobs and firm decisions about hiring, tasks, scheduling, insurance, and accommodation.
- Labor outcomes are the organizing objects: employment, hours, job choice, wages, productivity, mobility, firm behavior, and welfare.

Health as human capital and work capacity

Dimension	Labor interpretation	Typical proxy
Work capacity	ability to sustain hours and tasks	limitations, disability, function
Productivity	output conditional on working	performance, absence, demands
Risk	future impairment, exit, mortality	shocks, conditions, mortality risk
Feasible set	tolerable jobs and schedules	occupation, hours, mobility
Welfare	value of work net of pain, stress, risk	amenities, insurance, job quality

- Health shifts both the budget set and the feasible set.

Measurement choices

Measure	What it may capture	Main warning
Self-reported health	pain, fatigue, private functioning	reporting and justification bias
Diagnosis / biomarker Disability status	clinical status or severity health plus administrative criteria	may miss work capacity blends health and program rules
Hospitalization / shock Mental health scale Insurance / treatment	severe discrete event cognition, stress, persistence access and recovery margin	narrow margin latent and stigmatized jointly affects work and health

Different measures, different objects

- Self-reports can reveal private information about function, pain, fatigue, or latent conditions.
- Diagnoses and biomarkers can look objective but may miss severity, untreated illness, and task feasibility.
- Disability status is an institutional outcome, not pure work capacity.
- Mortality risk and acute shocks identify severe events, not routine chronic constraints.
- Insurance and treatment access can change health, reported health, labor supply, and mobility at the same time.

Reverse causality and dynamic selection

$$Y_{it} = \beta H_{it} + X'_{it}\Gamma + \alpha_i + \lambda_t + \varepsilon_{it}$$

- Reverse causality: work affects health through stress, injuries, income, insurance, schedules, and job conditions.
- Omitted SES: education, wealth, early-life resources, occupation, and local labor demand shape both health and work.
- Dynamic selection: healthier workers remain employed longer and sort into different firms, jobs, and schedules.
- Program rules: disability and insurance institutions can move labor supply independently of latent health.

Worker-side versus firm-side channels

Health channel	Worker-side margin	Firm-side margin
Reduced capacity	hours, exit, job downgrading	screening, reassignment, accommodation
Higher risk	safer jobs, retirement timing	insurance, safety investment
Treatment access	recovery, job lock, mobility	benefit design, job packages
Uncertainty	precautionary supply and saving	absenteeism and scheduling risk
Chronic burden	commuting and mobility limits	flexibility, monitoring, retention

- Reduced-form effects can mix supply, productivity, sorting, and employer response.

A field map for the course

Later object	Week 1 preparation
Disability and chronic conditions	separates health from program status and names the labor margin
Fertility, mortality, caregiving	treats shocks as lifecycle labor-allocation events
Mental health and productivity	makes latent conditions and welfare measurement explicit
Disease and environment	scales health from individual heterogeneity to market condition
Research synthesis	uses labor object - $\bar{z}$ health object - $\bar{z}$ design problem - $\bar{z}$ welfare object

- Primary anchor: Blundell, Britton, Costa Dias, and French on health measures and employment near retirement.
- Challenge anchor: Bound on self-reported versus objective health measures.
- Reproduce: compare reduced-form labor gradients across health measures.
- Diagnose: classify measurement objects and threats from reverse causality, SES, and selection.
- Transfer: move the logic to acute shocks, disability onset, mental health treatment, or insurance access.

Research design checklist

- ① Name the labor margin: employment, hours, wages, productivity, job choice, mobility, amenities, or welfare.
- ② Name the health object: self-report, diagnosis, biomarker, disability status, severe shock, mental health, insurance, or treatment.
- ③ State the timing and selection problem.
- ④ Separate worker-side and firm-side channels.
- ⑤ Say what welfare object is visible and what remains offstage.

- Week 1 builds the common language: health as capacity, risk, feasible set, measurement problem, and labor-market margin.
- Week 2 applies that language to disability, chronic conditions, job choice, accommodations, and program interaction.
- The next question: when does a health limitation become an employment constraint, a job-quality constraint, or an institutional response?

Takeaways

- Health is a labor object when it changes capacity, productivity, risk, feasible work, firm behavior, or welfare.
- Measurement is not a technical afterthought; it is one of the central open problems in the field.
- Reverse causality, omitted SES, and dynamic selection are baseline threats, not footnotes.
- Strong designs connect a health object to a labor margin, a channel, a counterfactual, and a welfare interpretation.